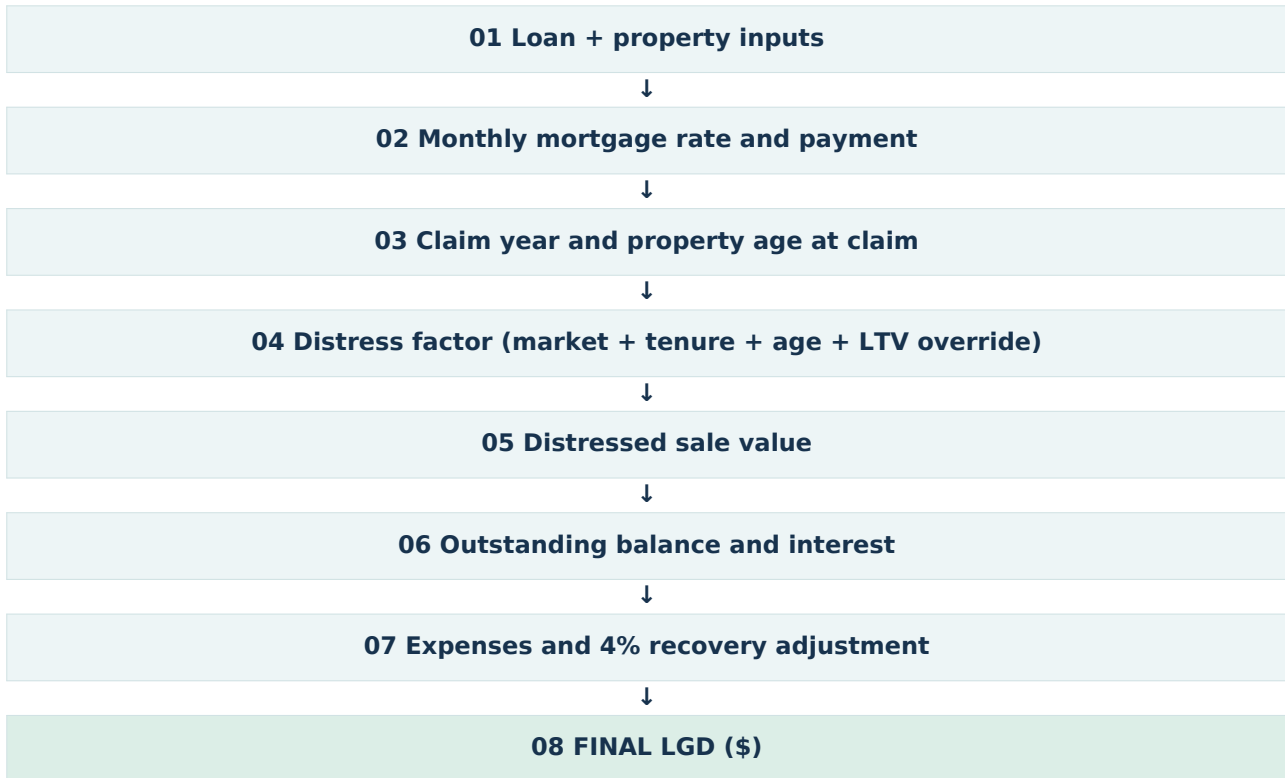


UPDATED LGD CALCULATION FLOW

Based on `lgd_calculator(3).py` and `lgd_config(1).yaml` | Amounts are monetary (\$), not an LGD percentage.

1. End-to-end calculation flow



2. Key parameters from the updated configuration

Parameter	Value
Month / settlement duration	36 months / 12 months
Recovery	4%
Distress: rural / urban condo / urban non-condo	35% / 30% / 25%
Low LTV: threshold / override	ACTLTV ≤ 80 / distress = 19%
Age adjustment	1 percentage point/year above 25; max 5 points
Property insurance rate	0.25%
Inflation scenario / expense base year	0.6 / 2026

3. Step-by-step formulas

Step 1 — Monthly interest rate and payment

$$r = (1 + R / 200)^{(1/6)} - 1$$

$$PMT = P \times r / [(1 - (1 + r)^{(-N)}) \times (1 + r)]$$

R = annual contract rate in percentage points; P = insured amount; N = amortization in months. At approximately zero interest, $PMT = P / N$.

Step 2 — Claim year and property age at claim

$$\text{StartYear} = \text{floor}(\text{EFFECT_SID} / 100)$$

$$\text{TotalClaimMonths} = 36 + (\text{EFFECT_SID} \bmod 100)$$

$$\text{ClaimYear} = \text{floor}(\text{StartYear} + \text{TotalClaimMonths} / 12)$$

$$\text{AgeAtClaim} = \text{PropertyAge} + \text{ClaimYear} - \text{StartYear}$$

Example: EFFECT_SID 202212 gives a claim year of 2026. If property age starts at 24, then age at claim is 28.

Step 3 — New rule-based distress factor

Baseline = 35% for rural (market code C or D); otherwise 30% for condo (tenure C); otherwise 25%. Rural takes priority.

$$\text{AgeAdj} = \min(\max(\text{AgeAtClaim} - 25, 0), 5) \times 0.01$$

$$\text{Distress} = 19\% \text{ if } \text{ACTLTV} \leq 80; \text{ otherwise } \text{Baseline} + \text{AgeAdj}$$

Example: urban condo, age 28, LTV 90 → 30% + 3% = 33%. If LTV is 75, use 19% instead.

Step 4 — Distressed property sale proceeds

$$\text{SaleValue} = \text{PropertyValue} \times \text{HPI_change} \times (1 - \text{Distress})$$

Example: $\$400,000 \times 1.02 \times (1 - 0.33) = \$273,360$.

Step 5 — Outstanding balance

$$D = \text{TotalClaimMonths} - 12; \text{ Growth} = (1 + r)^D$$

$$\text{Accumulated} = [(1 + r)^D - 1] / r \text{ (or } D \text{ for near-zero } r)$$

$$\text{OSB} = \max(0, P \times \text{Growth} - \text{PMT} \times \text{Accumulated} \times (1 + r))$$

Step 6 — Inflation and expense formulas

$$\text{Start-to-claim inflation factor} = \text{InflationIndex}(\text{ClaimYear}) / \text{InflationIndex}(\text{StartYear})$$

$$\text{Reference-year inflation factor} = \text{InflationIndex}(\text{ClaimYear}) / \text{InflationIndex}(2026)$$

Component	Formula implemented
Real estate commission	$\text{SaleValue} \times \text{RE_COMM_ADJ} / 100$
Taxes and condo fees	$(\text{MonthlyTax} + \text{CondoFee}) \times \text{StartToClaimInflation} \times 12$
Variable expenses	$\text{PROPERTY_VAR_EXPENSES_Igd} \times 12$
Fixed expenses	$(\text{Appraisal} + \text{LegalFees} + \text{MAINT}) \times \text{ReferenceYearInflation}$
Settlement interest	$\text{OSB} \times [(1 + r)^{12} - 1]$

$$\text{TotalExpenses} = \text{Commission} + \text{TaxesAndFees} + \text{Variable} + \text{Fixed} + \text{SettlementInterest}$$

Also computed but NOT included in TotalExpenses: $\text{PropertyInsurance} = \text{PropertyValue} \times 0.0025 / 12$; $\text{MaintenanceAdd} = \text{PropertyValue} \times \text{MAINT_ADD_Igd}$. These do not affect current LGD as the expense sum is written.

Step 7 — Final LGD

$$\text{GrossLoss} = \text{OSB} + \text{TotalExpenses} - \text{SaleValue}$$

$$\text{LGD} = \max(0, \text{GrossLoss} \times (1 - 0.04)) = \max(0, \text{GrossLoss} \times 0.96)$$

If required inputs or lookup values are missing, the calculator can return a missing LGD rather than a dollar amount.

4. Illustrative final calculation

Input/result	Amount
Outstanding balance (OSB)	\$350,000
Total expenses	\$35,000
Distressed sale value	\$273,360
Gross loss	\$111,640
Final LGD = \$111,640 × 0.96	\$107,174.40

5. Important implementation findings

These are issues in the uploaded code; the formulas above describe its intended logic, not a successfully executed run.

- `_add_lookups()` does not return `df`, so the following calculation receives `None`.
- `PROPERTY_VAR_EXPENSES_lgd` is referenced in `_calculate()` but not returned by `_load_expenses()`.
- `MAINT_ADD_lgd` is referenced during calculation but omitted from the expense lookup output.
- `ACTLTV` is used in `_prepare()` but is not listed in its required-column check.
- `property_insurance_lgd` and `property_maintenance_add_lgd` are calculated but are not part of `all_expenses_lgd`.
- The updated YAML repeats the columns: key. Both copies have the same contents, but duplication should be removed.
- The code may require corresponding columns such as `MAINT_ADD` and `INFLATION_FACTOR` in the expense file. Confirm their exact case and presence.

Source note: Uses only the uploaded updated calculator and YAML configuration; numerical examples are illustrative.